ALGORITHM ATLAS / CLASSIFICATION MAP

Two independent dimensions: HOW a model learns × WHAT problem it solves

LEARNING APPROACH

SUPERVISED

Linear regression

Decision tree · Random forest

Gradient boosting · SVM

CNN (often supervised)

UNSUPERVISED

K-means clustering

Principal component analysis

No target labels required

SELF-SUPERVISED / GENERATIVE

BERT: masked-token pretraining

GPT: next-token pretraining

GAN: adversarial objective

Diffusion: denoising objective

APPLICATION DOMAINS | The same model may appear in multiple columns

TABULAR DATA

Regression · trees · forests · boosting · SVM · K-means · PCA

COMPUTER VISION

CNN · PCA · GAN · diffusion

NATURAL LANGUAGE PROCESSING

SVM · BERT · GPT

GENERATIVE AI

GPT · GAN · diffusion

KEY DISTINCTION

Supervised / unsupervised = training signal. Vision / NLP / GenAI = task or output domain.

Self-supervised and adversarial/denoising objectives should not be forced into a binary label.

ALGORITHM ATLAS / METHODS 01–06

Predictive and exploratory models • concise mechanisms and applied examples

Linear Regression

Supervised | Tabular

HOW: Fits a line minimizing prediction errors.

USE: Forecast daily transit boardings.

Decision Tree

Supervised | Tabular

HOW: Splits records into rule-based branches.

USE: Classify stop productivity.

Random Forest

Supervised | Tabular

HOW: Averages many randomized decision trees.

USE: Predict paratransit trip demand.

Gradient Boosting

Supervised | Tabular

HOW: Sequential trees correct prior prediction errors.

USE: Estimate travel times.

Support Vector Machine

Supervised | Tabular · NLP

HOW: Finds a separating boundary with a wide margin.

USE: Classify service-request text.

K-means

Unsupervised | Tabular

HOW: Groups observations around learned centroids.

USE: Identify rider-demand clusters.

ALGORITHM ATLAS / METHODS 07–12

Representation learning, language, vision and generative model families

Principal Component Analysis

Unsupervised | Tabular · Vision

HOW: Projects data onto high-variance directions.

USE: Reduce correlated access indicators.

Convolutional Neural Network

Usually supervised | Vision

HOW: Learns local visual features using convolutions.

USE: Detect shelters in streetscape images.

BERT

Self-supervised → supervised | NLP

HOW: Pretrains on masked text; fine-tunes for tasks.

USE: Categorize rider feedback.

GPT-style Transformer

Self-supervised → aligned | NLP · Generative AI

HOW: Predicts successive tokens to generate text.

USE: Summarize planning reports.

Generative Adversarial Network

Adversarial training | Vision · Generative AI

HOW: Generator and discriminator compete.

USE: Create synthetic imagery.

Diffusion Model

Denoising objective | Vision · Generative AI

HOW: Learns to remove noise in iterative steps.

USE: Create streetscape concept images.